



Deer Browsing and Impact on Forest Development

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To cite this article: Christer Kaien (2006) Deer Browsing and Impact on Forest Development, Journal of Sustainable Forestry, 21:2-3, 53-64, DOI: [10.1300/J091v21n02_04](https://doi.org/10.1300/J091v21n02_04)

To link to this article: https://doi.org/10.1300/J091v21n02_04



Published online: 08 Sep 2008.



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LINKING BASICS AND MANAGEMENT

Implementation of sustainable forestry in southern Sweden calls for an improved knowledge base. Here we present results from selected studies of the south-Swedish forestry system. These range from improved understanding of the biogeophysical system to silvicultural aspects of sustainable forestry in southern Sweden.

Deer Browsing and Impact on Forest Development

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SUMMARY. Browsing by large herbivores can cause serious costs in forest production. The most important costs related to browsing damage

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The support for this study was provided by the Foundation for Strategic Environmental Research, MISTRA and the research programme Sustainable Forestry in Southern Sweden (SUFOR). Jonas Bergquist gave valuable comments on the manuscript.

[Haworth co-indexing entry note]: "Deer Browsing and Impact on Forest Development." Kalén, Christer. Co-published simultaneously in *Journal of Sustainable Forestry* (Food Products Press, an imprint of The Haworth Press, Inc.) Vol. 21, No. 2/3, 2005, pp. 53-64; and: *Sustainable Forestry in Southern Sweden: The SUFOR Research Project* (ed: Kristina Blennow, and Mats Niklasson) Food Products Press, an imprint of The Haworth Press, Inc., 2005, pp. 53-64. Single or multiple copies of this article are available for a fee from The Haworth Document Delivery Service [1-800-HAWORTH, 9:00 a.m. - 5:00 p.m. (EST)]. E-mail address: docdelivery@haworthpress.com].

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doi:10.1300/J091v21n02_04

are insufficient stand regeneration, volume losses and timber quality impairment. In Sweden, moose (*Alces alces*) and roe deer (*Capreolus capreolus*) are the most abundant deer species and therefore cause the majority of browsing damage. Balancing costs and benefits related to deer density is difficult since few data about the final losses in timber volume or quality impairment is available. Simulations of browsing and the impact on stand growth suggest that a minor or moderate browsing intensity has little effect on final harvest volume. The reason is that trees seem to recover some of the lost growth as they grow out of reach of the animals. Balancing between different uses of the forestry resource is complicated by our limited capability to predict the impact of different management regimes. [Article copies available for a fee from The Haworth Document Delivery Service: 1-800-HAWORTH. E-mail address: <docdelivery@haworthpress.com> Website: <<http://www.HaworthPress.com>> © 2005 by The Haworth Press, Inc. All rights reserved.]

KEYWORDS. Ungulates, silviculture, game, browsing damage, selectivity

INTRODUCTION

Sweden experienced a tremendous increase of moose and roe deer during the latter part of the 20th century (Cederlund and Markgren, 1987; Cederlund and Liberg, 1995). Several factors can explain this expansion such as introduction of modern silviculture, regulating game hunting, enclosure of domestic animals and a decline of natural predators (Kuiters et al., 1996). As a consequence of adopting modern silviculture practices during the 1950s and 60s, larger forest areas were felled, creating an excellent habitat for these animals. The carrying capacity for these animals therefore increased dramatically. The browsing damage caused by these animals has since then been identified as a threat to forest production and today the situation is alarming, at least when foresters are asked. In a recent enquiry, browsing damage was ranked as the most serious threat to forest development (Blennow and Sallnäs, 2002). Recreational hunting is, on the other hand, an important leisure activity for about 350,000 Swedes. Today, roughly 100,000 moose and 150,000 roe deer are harvested annually (Anon, 2003), where roughly 500 million SEK can be accredited to the meat value alone (Jämttjärn, 1996). Hence, both costs and benefits need to be evaluated to find an appropriate balance between forest damage and recreational hunting.

Also, the present moose density potentially affects forest biodiversity. For example, high densities of large browsing herbivores hamper the development of deciduous tree species such as aspen (*Populus tremula* L.) and rowan (*Sorbus aucuparia* L.) (Angelstam et al., 2000; Ericsson et al., 2001), species that are important for the existence of many organisms.

This paper discusses the results of a project focused on modeling browsing intensity and the related impact on stand development. A specific objective is to draw some implications for management of deer and forests.

DIET PREFERENCES

Although both moose and roe deer are considered generalist feeders, they prefer to browse on certain plant species (e.g., Bergström and Hjeljord, 1987). The summer diet is composed of herbs, leaves and dwarf shrubs and the winter diet of tree twigs and, if the snow cover is limited, dwarf shrubs. Since twigs on trees mainly are used during winter this is the period when browsing damage occurs. These animals are, however, selective towards different tree species also in winter and trees are therefore not consumed in proportion to occurrence (Bergström and Hjeljord, 1987). Norway spruce (*Picea abies* L.) is, for example, usually avoided by moose. Due to a combination of abundance and palatability, Scots pine (*Pinus sylvestris* L.) and birch (*Betula pendula* Roth. and *Betula pubescens* Ehrh.) make up the greatest part of the winter diet of moose in Sweden. Bergström and Hjeljord (1987) found that moose in Europe rank the most common tree species in the following order of preference: Rowan > Willows (*Salix* L.) > aspen > Common juniper (*Juniperus communis* L.) > Birch > Scots pine > Alder (*Alnus glutinosa* L.) > Norway spruce.

The preference for certain tree species causes browsing damage to be unevenly distributed on forest stands (Hörnberg, 1995). One cannot therefore assume that all (young) stands have an equal risk of being subjected to browsing damage. Instead, pine and birch stands will suffer higher browsing damage than spruce stands. Due to its lower susceptibility for browsing damage (Lundén, 1996) as well as lower incidence of browsing attacks, this strengthens the present trend of managers preferring stand regeneration with Norway spruce. As the area covered by spruce increases, one should therefore realise that relative browsing damage is likely to increase in areas covered with other tree

species, given that the density of deer is held constant through hunting. Thus, there are two alternative ways to decrease the relative browsing damage, either by lowering the animal density or by increasing the availability of preferred food.

THE CONSEQUENCE OF MOBILITY

Both moose and roe deer are highly mobile and use a large area for finding forage. Although not territorial, they are philopatric, which means that they are usually found within a given home range. The size of the home range decreases if resource availability and overall density of animals increase. Although resource availability can explain parts of the spatial distribution on a landscape level, other factors such as risk for predation, human settlement, food accessibility and site productivity also determine where animals are located and where they forage. Little is known to what extent these factors determine spatial distribution. For management it means that we have poor capacity to predict where browsing damage will occur. Hence, on a smaller landscape level, stands with similar availability of food can receive very different browsing damage (Hörnberg, 1995). Regulating population density efficiently through hunting to purposefully limit damage to forests therefore requires coordination on a larger area than a typical private forest estate in southern Sweden (ca. 30-50 ha).

FORAGE AVAILABILITY

In theory, increasing the availability of edible food will decrease the relative browsing damage. However, for this to be efficient it must be made on a large scale. Otherwise one may only attract more animals to sites rich in resources. Studies have shown that increasing available forage by delaying, for example, cleaning of a stand, has a minor effect on reducing browsing damage on the economically important trees (Härkönen, 1998). The reason is probably that the stand attracts more animals or that animals' residence time in the stand is prolonged. The moose is in winter known to "park" in young pine stands, causing strong local damage. The reason for this is probably that an even-aged monoculture offers a vast amount of food and there is no point in spending energy by walking, especially if snow cover is extensive. A young pine stand may hold about 1-3 tonnes of twig biomass per ha (Kalén and

Bergquist, 2004), which is more than required by a single moose during a winter season. A planted monoculture has an excessive amount of food during the first 25 years of the rotation period, which for pine is about 100 years in southern Sweden. After trees escape the foraging window virtually no food is available until the stand is felled and regeneration takes place. Compared to a natural forest this cyclic pattern is very different. Except for occasional disturbance factors such as fire, insect outbreaks and storm fellings, a natural forest has in general very little food to offer (Reimoser and Gossow, 1996). As trees become higher the risk for the leader being browsed decreases since it is out of reach for the animals. Browsing of lateral shoots or shoots located lower in the crown will have little effect on growth or timber quality impairment. One way to increase the availability of food in the landscape, without increasing the risk of experiencing extensive costs from browsing damage, is to avoid cleaning and delaying the thinning. It is known, however, that the animals do not prefer to browse in lower parts of the crown (Hjeljord et al., 1990; Gill and Beardall, 2001), so although food is available in such stands the animals need not visit them.

IMPACT ON PLANT PERFORMANCE

The impact of herbivory on plants is determined by its intensity (Jackson, 1980), its nature (i.e., the parts of the plant that are attacked) and the ability of the plants to compensate for herbivory (Taylor and Bardner, 1968). The response to herbivory varies both between individual plants and between plant species (McNaughton, 1983; Crawley, 1983; Crawley, 1997). Plant responses to browsing can lead to a decrease in growth, an increase in growth or no effect at all (compensatory growth) (McNaughton, 1983). Thus, in the short-term, browsing will not always have a negative impact on plant productivity. However, compensatory growth is mediated by the use of stored carbohydrates and eventually, after repeatedly being subjected to browsing, plants' capacity to recover lost growth will decline (Crawley, 1983). This may also explain why the risk of mortal damage is higher for seedlings than for larger trees. The impact is therefore also dependent on the timing of attack relative to plant development and the history of attacks. Plants' responses to browsing are also influenced by site productivity (Edenius, 1993; Edenius et al., 1995). In clipping studies reported by Danell et al. (1997), clipped birch produced larger shoots, with larger leaves, than

unclipped birch. Browsing will therefore not always have a negative impact on plant productivity.

Simulated clipping of young spruce trees shows a clear compensatory growth leading to marginal losses of productivity some years after browsing has ceased as compared to the control (Bergquist et al., 2003). However, tree species used in commercial forestry differ significantly in the response to browsing (Bergquist et al., 2003). At present we have little knowledge on the final loss in harvested volume as a consequence of browsing but simulation studies suggest that minor or moderate browsing intensity has little effect on the final harvest volume (Kalén, and Bergquist, 2004). However, more empirical studies on the recovery from browsing are needed to be able to evaluate the long term consequences on stand growth.

ESTIMATING BROWSING IMPACT—A SIMPLE MODEL

There are no empirical data on the degree of browsing damage that can be attributed to moose and roe deer, respectively. Obviously, a single moose would have a greater impact than a single roe deer since it is about ten times larger and therefore requires more food. One could therefore draw the conclusion that given equal density of animals, the moose would render ten times more damage as compared to roe deer. However, there are other factors that may influence the degree of damage. For example, the moose, which can use taller trees, may therefore use a forest stand for a longer time compared to roe deer.

With empirical data from Kalén and Bergquist (2004), a function describing available food in a pine stand was used to describe the increase of food availability as the stand develops, and its decline as the trees escape the foraging window (Figure 1). In this example, the foraging windows were set to 0.5–2.5 and 0–1.5 metres for moose and roe deer, respectively. The total biomass of twigs up to 4 mm in diameter located within the foraging window is considered available for moose. To take into account the lower bite diameter of roe deer, the available biomass for the species was divided by two, corresponding to a bite diameter of approximately 2 mm. The resulting changes in food availability during the development of a pine stand (3,000 stems per ha) are shown in Figure 2.

It is clear from Figure 2 that moose can utilise a much larger fraction of the pine biomass within the stand than roe deer. The average amounts of food available per year are 900 kg over a 26-year period for moose and 180 kg over a 20-year period for roe deer. Assuming a rotation pe-

FIGURE 1. The available forage during stand development is estimated through relationships between stand age and tree height as well as stand age and stem diameter. Stem diameter determines the crown's total twig biomass. Physical limitations of moose and roe deer determine the amount of twig biomass available within the foraging window.

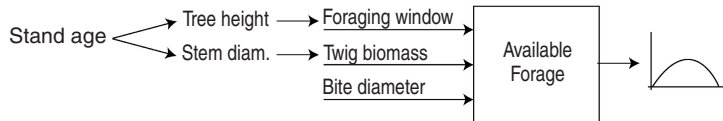
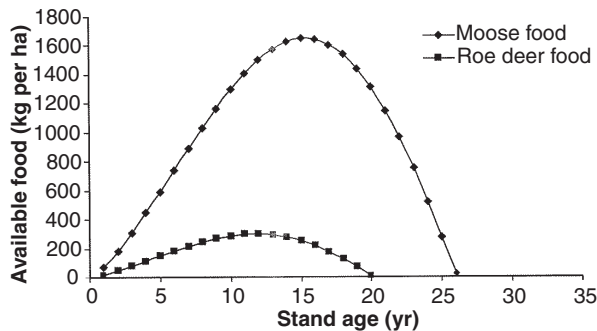


FIGURE 2. Development of food availability during the first 35 years of stand growth.



riod (r) of 100 years and an equal age distribution (i.e., all age classes in the landscape are represented in equal amounts), we can estimate the average amount of food available per ha (f) at the forested landscape level. This can be done by dividing the total (cumulative) amounts of food available (a) for the two deer species (i), shown in Figure 1, by the rotation period.

$$f = a_i/r$$

This implies that $25,000/100 = 250$ kg per ha is available for moose and $3,600/100 = 36$ kg per ha for roe deer. Hence, in a landscape forested by Scots pine there is about 250 and 36 kg of food available per ha for moose and roe deer, respectively. It is important to note that food available in the field layer is not considered here.

A moose requires about 1,000 kg of twig biomass (c) to survive a winter (approximately 6 kg per day during 150 days). Thus, 4 ha (c/f) are needed to meet the demands of a single moose. Assuming that a roe deer would require 100 kg of twig biomass per winter, 3 ha are needed per individual. Now, this implies 100% browsing efficiency and an animal density of 25 or 33 individuals (moose and roe deer, respectively) per km² (250 and 330 per 1,000 ha), which of course is way over the long-term carrying capacity. To receive 2% yearly damage rate on young pines, at present the existing compromise between hunter and foresters, the model implies a density of 0.5 or 0.7 individuals per km² (moose and roe deer, respectively).

Assume now that the values used to estimate this density have an uncertainty of, let's say, $\pm 10\%$. Thus, instead of defining the daily consumption of moose to 6 kg we could use either 5.4 or 6.6 kg. A similar procedure for the other parameters (number of winter days, availability and rotation period) suggest the uncertainty interval to be between 0.3-1.1 moose per km² in order to receive 2% yearly browsing damage. This includes the present estimate of 1 moose per km² used by hunters and foresters.

Having obtained these figures, we are able to estimate the amount of game meat produced per ha by dividing the slaughter weight by the area required. For moose, this gives about 125 kg of meat and, hence, about 30 kg (125/4) per ha. The corresponding figure for a roe deer is 4 kg (12.5/3) per ha assuming that it provides only a tenth as much meat as a moose. Hence, the meat production per ha is about seven times larger for moose than for roe deer. If we were only interested in maximising meat production this simple analysis suggests that moose is a better option than roe deer. However, relating these figures to browsing damage is more complex. A single moose can potentially consume seven times the amount of biomass per ha compared to roe deer. In addition, a moose can potentially retard the trees for a longer period than roe deer. In this respect, the moose would therefore have a larger impact. However, per kilo of meat, damage caused by moose will be more scattered than damage by roe deer, and the browsing intensity per tree will be lower. It is important to note that this example cannot be used to estimate carrying capacity since animals rarely consume all of the biomass in a stand before leaving for a more profitable site.

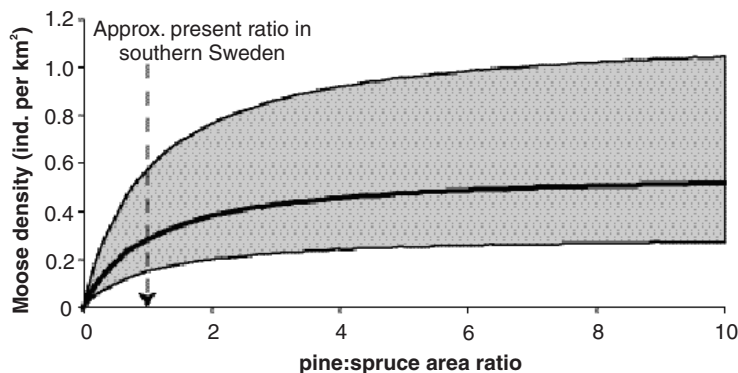
It is also important to point out that the damage level is dependent on both age structure of trees and the pine:spruce area ratio. In the above analysis we have considered a landscape of Scots pine only. For south-

ern Sweden the present pine:spruce area ratio is about 1. Since spruce usually is avoided by these animals, a decrease in this ratio will increase the damage in the pine forest, given of course that the animal density is held constant. As seen in Figure 3, increasing the ratio will make possible a higher density of moose without increasing the damage level above 2%.

DISCUSSION

The complexity of the behavioural pattern of large herbivores and the ambiguity of the response to browsing in plants make difficult any attempt to predict the impact of browsing on a given forest stand. On a large scale we can with fair certainty conclude that more herbivores will result in higher browsing intensity, given that the food availability is held constant. However, we have poor capacity to predict how browsing will be distributed in the landscape. Young stands of the preferred species (Scots pine and birch) will receive a higher intensity, but even here the browsing will be unevenly distributed due to differences in, e.g., plant productivity (Edenius, 1993), previous browsing (Heikkilä, 1991; Danell et al., 2003), and stand height (Bergström and Danell, 1987; Faber and Lavsund, 1999; Palmer and Truscott, 2003).

FIGURE 3. The pine:spruce area determine the density of animals required to cause 2% yearly browsing damage. The grey area represents a $\pm 10\%$ uncertainty interval for the parameters used in the calculation.



On a landscape level we must conclude that an increase in spruce stands will increase browsing damage on the preferred trees. Thus, while planting tree species avoided by the animals (such as Norway spruce) may reduce the impact on the specific stand, it will increase the intensity somewhere else. Hence, given that the herbivore population is held constant, one can view the browsing pressure as communicating buckets: reducing the food in one end will increase the browsing pressure somewhere else. When property rights in Swedish forestry is considered, we can therefore view the issue as a case of “tragedy of the commons” (Hardin, 1968). While an individual forest owner will reap the benefits (i.e., less browsing impact) by planting spruce, the cost will be distributed amongst other forests in the landscape. This problem has no solution other than communication and the setting of rules among the foresters.

Depending on the definition of sustainability, large herbivores have the potential to greatly influence parameters for sustainable forestry. Social sustainability is promoted if the conflict between forest managers and game hunters is alleviated. Sweden has well developed institutional capacity for negotiation between stakeholders for setting harvest quotas. However, since both moose and roe deer are very mobile animals, it is likely that the present harvest strategies are managed at a too small spatial scale. Setting harvest quotas on a small spatial scale may be an inefficient way of controlling browsing damage, as the presence of animals is influenced by neighbouring hunting concession areas. It is difficult to scientifically justify the proper scale at which animal density should be regulated, but it would probably require between 4-100 km².

The risk that individual plant species will go extinct due to high densities of herbivores is probably low on a national level but can be locally important. More troublesome is the low variety of commercially used tree species, partly as a consequence of herbivore selectiveness in feeding. This indirect effect of browsing has probably a larger impact on the plant species composition than browsing itself and therefore also on biological diversity.

The society needs to account for multiple use of forests to satisfy different stakeholders using forests as a resource. An improved system for valuing game hunting as well as monitoring the cost of browsing damage is needed for such management. However, to enable use of such instruments in a predictive manner we need enhanced capability to estimate spatial distribution of animals, their foraging behaviour, and the response of trees subjected to attack from these herbivores. In addition, tools for predicting browsing damage under different silviculture regimes needs development. Such tools must also be cost-efficient and

should therefore not require costly parameterisation. The simple tool presented in this paper is in this respect cheap since it requires less parameterisation. However, the capacity of such a simple model to predict browsing damage with desired accuracy still remains to be tested. For example, this model does not consider the capacity of trees to recover and repair browsing damage. Hence, there is often a trade-off between robust models that are costly to parameterise and complicated to use, and simple models with less predictive capacity yet requiring less parameterisation.

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